

		hence, $F_{\text{air resistance}} = \text{weight} = 50 (10) = \underline{500 \text{ N}}$.
3	C	vertically, $v_y^2 = u_y^2 + 2 a_y s_y$ $v_y^2 = (-50.0 \sin 30.0^\circ)^2 + 2 (9.81) (30)$ $v_y = 34.837 \text{ m s}^{-1}$ horizontally, $v_x = (50.0 \cos 30.0^\circ) = 43.301 \text{ m s}^{-1}$ $\tan \Theta = 43.301 / 34.837$ $\Theta = 51.2^\circ$
4	A	$(0.020)(8) - (0.010)(7) = (0.020 + 0.010)v_f$